

Dead Sea Shoreline Recession Analysis Using GIS Remote Sensing (1984 to 2021)

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Abstract

The Dead Sea, sitting at the border of Israel and Jordan in the Middle East, was studied in order to analyze the surface area change of the Dead Sea from 1984 to 2021 using the remote sensing data and geographic information system tools. The results obtained from the study has shown an overall decline of approximately 20.1 Km² of area was experienced by Dead Sea over 37 years. When zooming into the shrinkage more it was observed that the southern basin has particularly shown a significant decline. Monthly water classification raster data from JRC Global Surface Water dataset were used to process in ArcGIS Pro and the raster to polygon tool was used for conversion and geometric calculation tools were used for quantifying annual water extent across six time periods. The findings have confirmed a long-term trend of shrinkage which is driven by multiple factors for example reduced freshwater inflow and increased industrial extraction, and the outcome and the reasons are actually consistent with previously published research on Dead Sea hydrology.

Introduction

The Dead Sea, located at the border between Israel, Jordan, and the Palestinian territories, is one of the most hypersaline water bodies on Earth and one of the most studied examples of terminal lake recession in the world. Its surface lies approximately 430 meters below sea level, making it the lowest point on the surface of the Earth (Oren, 2010). Over recent decades, the Dead Sea has experienced significant and well-documented water loss primarily due to human interference with its primary source, the Jordan River, and large-scale potash and mineral extraction by industrial operations on both the Israeli and Jordanian sides of the lake (Lensky et al., 2005).

Remote sensing and GIS technologies have become essential tools for monitoring changes in surface water bodies at large spatial scales and across long time periods. Satellite imagery analysis

allows researchers to detect and quantify changes in water extent with high spatial resolution and temporal consistency (Pekel et al., 2016). The JRC Global Surface Water dataset, developed by the European Commission Joint Research Centre, provides a comprehensive archive of global surface water changes derived from Landsat imagery spanning from 1984 to the present day, making it an ideal resource for long-term lake monitoring studies (Pekel et al., 2016).

Previous studies have documented the rapid decline of the Dead Sea using various methodologies including ground-based measurements, photogrammetry, and satellite remote sensing. Gavrieli et al. (2005) documented water level drops of approximately one meter per year during the late twentieth century. More recent work by Salameh et al. (2019) demonstrated that the southern basin of the Dead Sea had been almost entirely converted to evaporation ponds operated by the Dead Sea Works industrial company. Understanding the spatial extent of this recession is critical for water resource management, regional ecological planning, and informing international water-sharing agreements between riparian states.

This project applies GIS-based spatial analysis techniques to map and quantify the surface area change of the Dead Sea using freely available remote sensing data. The study covers six time periods spanning 1984 to 2021 and uses vector-based polygon area calculation methods in ArcGIS Pro to produce quantitative estimates of water body extent at each time step. The results are presented both spatially through cartographic outputs and numerically through area statistics tables.

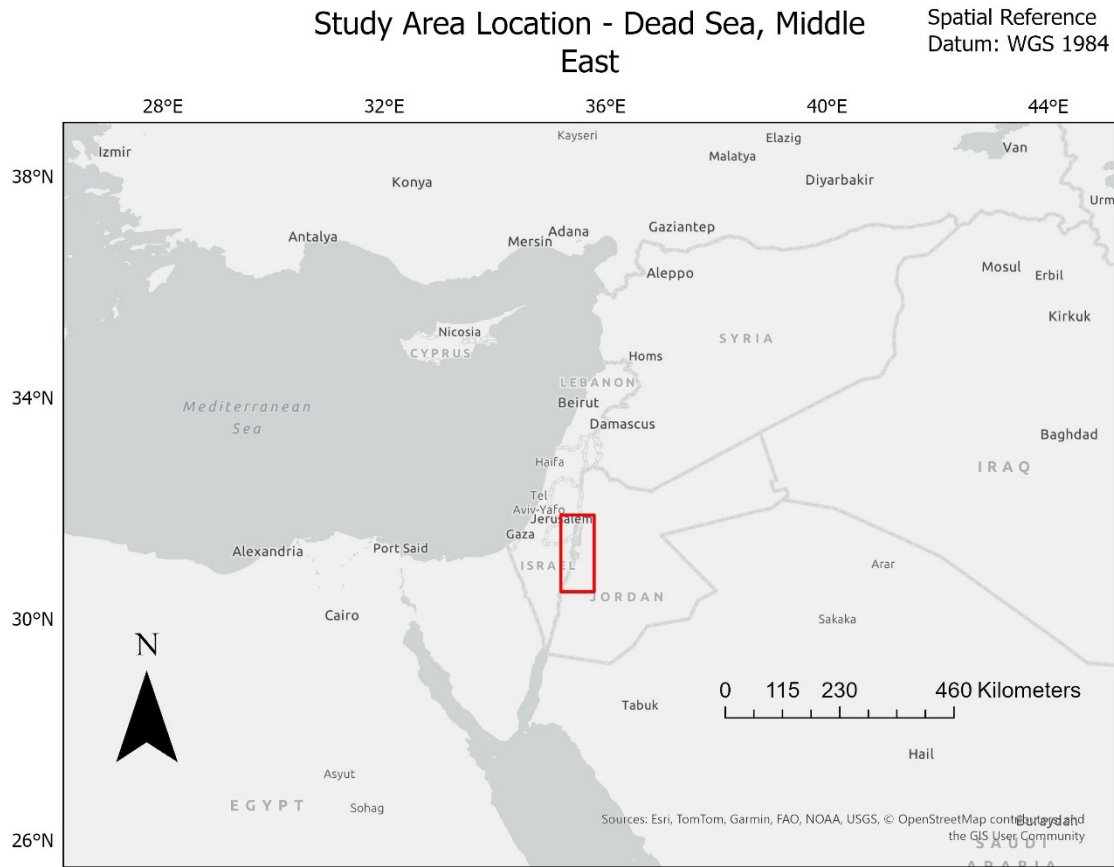


Figure 1: Study Area Location. The red rectangle indicates the Dead Sea study area located between Israel and Jordan in the Middle East. Coordinate system: WGS 1984.

Methods

Surface water extent data for the Dead Sea region were obtained from the JRC Global Surface Water Monthly History dataset, version 1.4, accessed through the Google Earth Engine cloud computing platform (Pekel et al., 2016). This dataset provides monthly global surface water classifications derived from Landsat 5, 7, and 8 imagery at 30-meter spatial resolution. For each study year, the August monthly image was selected to minimize seasonal variability and cloud cover, which is consistent with best practices for arid region water body monitoring (Ji et al., 2009).

Six study years were selected to capture both early baseline conditions and recent changes: 1984, 1990, 1998, 2000, 2015, and 2021. The Dead Sea study area was defined using a geographic bounding box of 35.2 degrees East to 35.8 degrees East longitude and 30.5 degrees North to 31.9 degrees North latitude. For each year, pixels classified as water were extracted using a waterClass

threshold of greater than or equal to 1, capturing both permanent and seasonal water classifications. Each annual water raster was exported from Google Earth Engine as a GeoTIFF file at 30-meter resolution.

All spatial analysis was conducted in ArcGIS Pro version 3.x. The six raster datasets were imported into a file geodatabase named FinalprojectASH.gdb along with their corresponding vector polygon outputs. Each raster was converted to a polygon feature class using the Raster to Polygon geoprocessing tool with polygon simplification enabled. A new field named Area km2 was added to each polygon attribute table, and the Calculate Geometry tool was applied to compute the geodesic surface area of each polygon in square kilometers. Only polygons with gridcode equal to 1 were selected to exclude non-water areas from the total area calculation. The sum of all water polygon areas for each year was used as the total surface area estimate for the Dead Sea at that time period.

Cartographic outputs were produced in ArcGIS Pro using the Layout view. The primary output map displays all six shoreline polygons overlaid on a light gray canvas basemap, styled with distinct colors for each year to allow visual comparison of shoreline positions. Map elements including a north arrow, scale bar, graticule with latitude and longitude labels, and legend were added according to standard cartographic requirements.

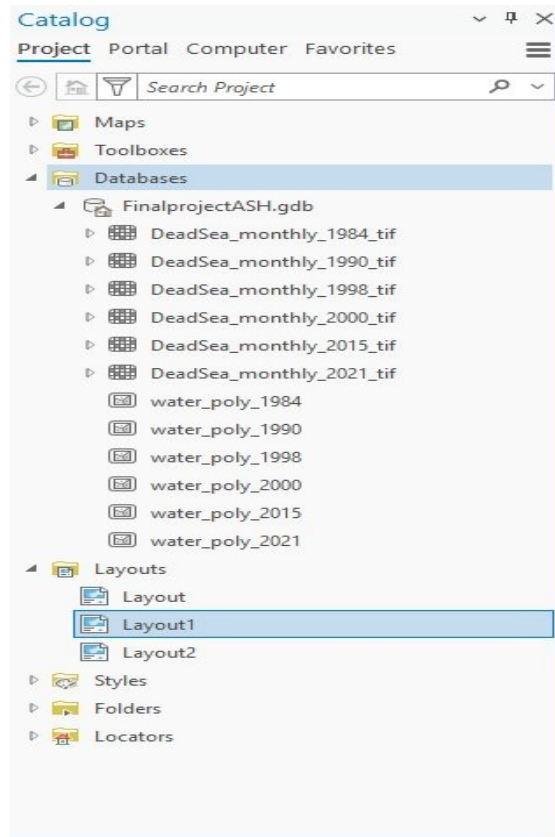


Figure 2: File Geodatabase Structure (FinalprojectASH.gdb) showing six raster datasets and six vector polygon feature classes used in the analysis.

Results

The spatial analysis produced surface area estimates for the Dead Sea across six time periods from 1984 to 2021. The results are summarized in Table 1 and Figure 4. The total surface area ranged from a minimum of 861.4 km² in 2021 to a maximum of 921.3 km² in 1998. The baseline area in 1984 was calculated as 881.5 km².

Table 1: Dead Sea Surface Area by Year

Year	Surface Area (km ²)	Change from 1984 (km ²)
1984	881.5	Baseline
1990	893.0	+11.5
1998	921.3	+39.8
2000	914.1	+32.6
2015	880.9	-0.6

2021	861.4	-20.1
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The results indicate a non-linear pattern of change over the study period. Between 1984 and 1998 the surface area increased by approximately 39.8 km², reaching its peak value of 921.3 km². This period of apparent increase may reflect inter-annual variability in precipitation and Jordan River inflow. From 1998 onward the trend reversed, with the surface area declining steadily to 880.9 km² in 2015 and 861.4 km² in 2021. The net change from 1984 to 2021 represents a loss of 20.1 km², equivalent to approximately 2.3 percent of the 1984 baseline area.

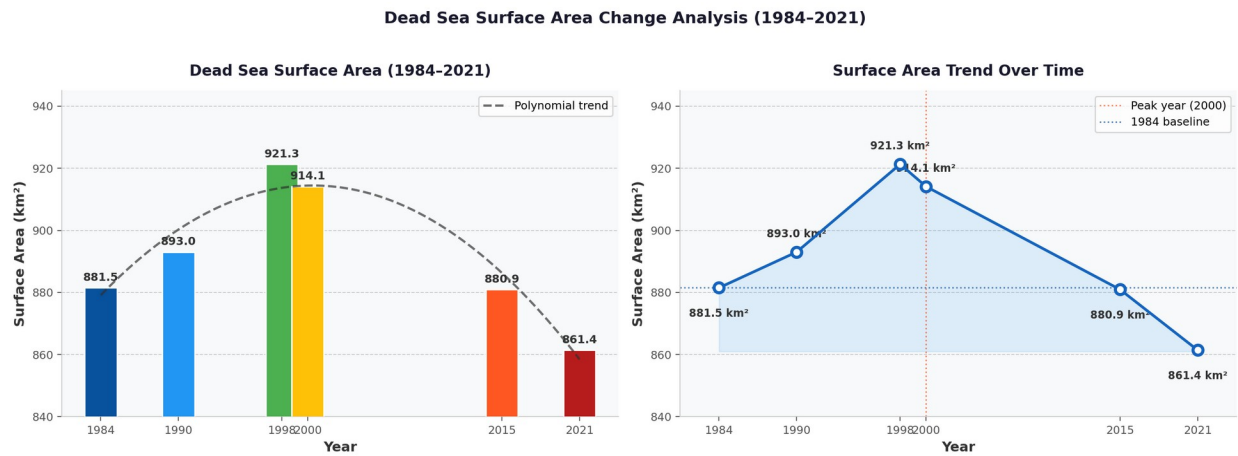
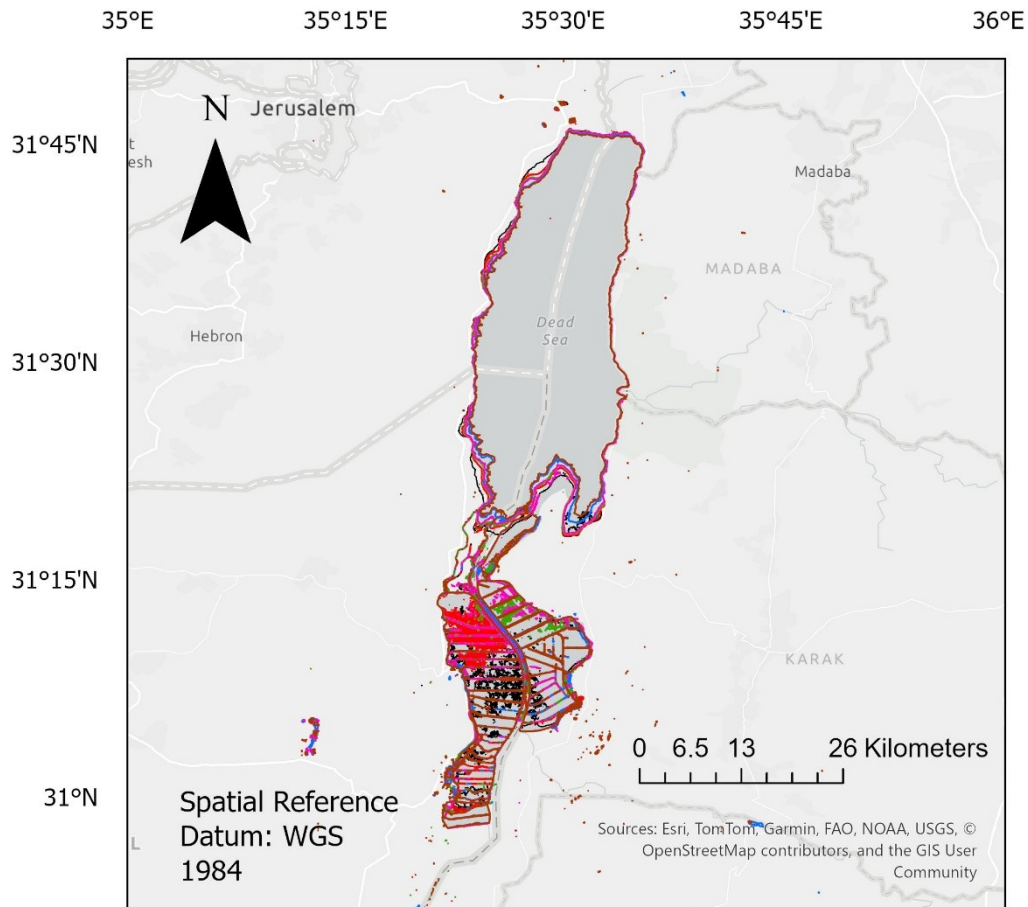


Figure 4: Left panel shows Dead Sea surface area (km²) per year with a polynomial trend line. Right panel shows the area trend over time with the 1984 baseline and peak year reference lines. Both charts confirm the post-2000 declining trend.

The spatial pattern of change is clearly visible in Figure 3, which shows the overlaid shoreline polygons for all six study years. The most pronounced recession is observed in the southern basin of the Dead Sea, where industrial evaporation ponds have replaced the natural water body. The northern basin shows more modest shoreline retreat concentrated along the eastern and southern margins. The 2021 shoreline is noticeably smaller than the 1984 extent across both basins.

Dead Sea Shoreline Recession Analysis (1984–2021)



Dead Sea Water Extent (1984-2021)



Figure 3: Dead Sea Shoreline Recession Analysis (1984 to 2021). Overlaid polygon extents show water body boundaries for six study years on a light gray canvas basemap. Each color represents a different year. Coordinate system: WGS 1984.

Conclusions

This study successfully applied GIS remote sensing techniques to document and quantify the surface area change of the Dead Sea over a 37-year period from 1984 to 2021. The results confirm a net decline of 20.1 km² in total water surface area, with the most significant losses occurring in the southern basin after 1998. These findings are consistent with previously published studies documenting Dead Sea recession driven by reduced Jordan River inflow and industrial water extraction (Gavrieli et al., 2005; Salameh et al., 2019).

The methodology employed in this study demonstrates the effectiveness of freely available remote sensing datasets for monitoring long-term changes in surface water bodies. The JRC Global Surface Water dataset provided a consistent and reliable source of annual water classification data across multiple decades, enabling direct comparison of water extent at different time periods. The use of ArcGIS Pro for raster to polygon conversion and geodesic area calculation produced quantitative results that are well-suited for reporting and comparison with other studies.

There are some limitations to this analysis that should be considered. The selection of August monthly imagery for each study year introduces potential bias from inter-annual precipitation variability and does not capture the full seasonal range of water extent. Additionally, the JRC water classification algorithm may underestimate water extent in hypersaline environments due to the unusual spectral reflectance properties of the Dead Sea compared to freshwater bodies (Oren, 2010). Future work could address these limitations by using multi-temporal composites rather than single monthly images and by applying specialized spectral indices calibrated for saline water bodies.

The results of this study have broader implications for regional water management and environmental policy. The continued decline of the Dead Sea represents a significant ecological and cultural loss for the region, and quantitative documentation of its spatial extent is essential for informing ongoing negotiations between Israel, Jordan, and the Palestinian Authority regarding the proposed Red Sea to Dead Sea water conveyance project. GIS-based monitoring approaches like the one demonstrated here provide a cost-effective and scalable method for tracking progress toward restoration goals and informing adaptive management decisions.

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